

Table 1: Collision cost, cost_limit=0: SafeDreamer (100k steps) vs MACPO (3000k steps).

Algorithm	Scenario	Score $\uparrow$	Cost $\downarrow$	Winrate $\uparrow$	Steps	Cost Limit
MACPO	2s_vs_1sc	19.73 ± 0.21	0.00 ± 0.00	0.94 ± 0.03	3000k	0
Safe Dreamers	2s_vs_1sc	19.86 ± 0.22	0.00 ± 0.00	0.95 ± 0.03	100k	0
MACPO	3m	19.31 ± 0.84	0.00 ± 0.00	0.95 ± 0.06	3000k	0
Safe Dreamers	3m	19.03 ± 0.91	0.02 ± 0.02	0.92 ± 0.07	100k	0
MACPO	2s3z	14.65 ± 0.78	0.01 ± 0.01	0.24 ± 0.11	3000k	0
Safe Dreamers	2s3z	18.72 ± 0.44	0.16 ± 0.11	0.79 ± 0.07	100k	0
MACPO	3s_vs_3z	13.18 ± 1.83	0.00 ± 0.00	0.15 ± 0.16	3000k	0
Safe Dreamers	3s_vs_3z	19.89 ± 0.05	0.02 ± 0.03	0.97 ± 0.01	100k	0
MACPO	3s_vs_4z	7.28 ± 0.20	0.00 ± 0.00	0.00 ± 0.00	3000k	0
Safe Dreamers	3s_vs_4z	15.49 ± 3.71	0.26 ± 0.23	0.06 ± 0.09	100k	0
MACPO	3s_vs_5z	6.35 ± 0.31	0.01 ± 0.01	0.00 ± 0.00	3000k	0
Safe Dreamers	3s_vs_5z	13.72 ± 2.97	0.08 ± 0.07	0.01 ± 0.03	100k	0
MACPO	8m	15.79 ± 2.00	0.39 ± 0.05	0.54 ± 0.17	3000k	0
Safe Dreamers	8m	17.49 ± 0.43	0.58 ± 0.06	0.71 ± 0.05	100k	0
MACPO	MMM	9.78 ± 0.46	1.15 ± 0.20	0.00 ± 0.00	3000k	0
Safe Dreamers	MMM	15.12 ± 0.17	1.20 ± 0.52	0.39 ± 0.03	100k	0
MACPO	3s5z_vs_3s6z	8.71 ± 0.12	0.03 ± 0.03	0.00 ± 0.00	3000k	0
Safe Dreamers	3s5z_vs_3s6z	10.65 ± 0.82	0.86 ± 0.08	0.01 ± 0.01	100k	0
MACPO	bane_vs_bane	—	—	—	3000k	0
Safe Dreamers	bane_vs_bane	18.20 ± 0.59	0.15 ± 0.13	0.57 ± 0.10	100k	0